
Step-by-Step Numerical Calculations of Novel NDVI Analysis Terms

*Scale Correction · Anomaly · STL Decomposition · Mann–Kendall Test ·
CVSI*

Study Pixel: Pauri Garhwal Forest, Uttarakhand, India

Coordinates: 30.15°N, 78.78°E Elevation: $\approx$ 1,420 m

Forest type: Sub-tropical broadleaved humid forest

Data source: MOD13A3 v061 (1 km monthly NDVI)

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Reference: Biswas et al. (2025), *Environ. Sci. Pollut. Res.*,
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1 Reference Pixel and Input Data

All calculations in this document are performed for a **single spatial pixel** located in the Pauri Garhwal district of Uttarakhand — a confirmed high fire-risk zone identified by ? (Fig. 6, 10 therein).

Attribute	Value
Location	Pauri Garhwal, Uttarakhand, India
Coordinates	30.15°N, 78.78°E
Pixel ID	Row 45, Column 127 (in India 1 km grid)
Forest type	Sub-tropical broadleaved humid
Fire event	MODIS FIRMS confirmed, April 2016

Table 1 gives the raw integer NDVI values (**raw**) from MOD13A3 for this pixel over 12 representative months (January 2015 – December 2016), used in all calculations below.

Table 1: Raw MOD13A3 integer NDVI values and QA flags for the reference pixel (2015–2016)

Year	Month	Raw value	QA flag q	Scaled NDVI	QA Status
2015	Jan	3 750	0	0.3750	Good
2015	Feb	3 620	0	0.3620	Good
2015	Mar	3 480	0	0.3480	Good
2015	Apr	3 290	0	0.3290	Good
2015	May	3 150	1	0.3150	Marginal (usable)
2015	Jun	5 700	0	0.5700	Good
2015	Jul	7 610	0	0.7610	Good
2015	Aug	7 820	0	0.7820	Good
2015	Sep	7 200	0	0.7200	Good
2015	Oct	5 400	0	0.5400	Good
2015	Nov	4 100	0	0.4100	Good
2015	Dec	3 900	0	0.3900	Good
2016	Jan	3 620	0	0.3620	Good
2016	Feb	3 450	0	0.3450	Good
2016	Mar	3 300	0	0.3300	Good
2016	Apr	−3000	−1	NaN	FILL (fire scar)
2016	May	3 020	0	0.3020	Good
2016	Jun	5 500	3	NaN	Cloudy—excluded
2016	Jul	7 410	0	0.7410	Good
2016	Aug	7 700	0	0.7700	Good

2 Step 1 — Scale Correction and QA Filtering

Step 1A — Scale Correction (Equation F1)

Formula:

$$\text{NDVI}_{ij}^{(t)} = \begin{cases} \text{raw}_{ij}^{(t)} \times 10^{-4} & \text{if } -2000 \leq \text{raw}_{ij}^{(t)} \leq 10000 \\ \text{NaN} & \text{otherwise (fill or out-of-range)} \end{cases} \quad (\text{F1})$$

Worked calculations for January 2015 through March 2016:

$$\text{Jan-2015 : } \text{NDVI} = 3,750 \times 10^{-4} = \mathbf{0.3750} \quad (1)$$

$$\text{Mar-2015 : } \text{NDVI} = 3,480 \times 10^{-4} = \mathbf{0.3480} \quad (2)$$

$$\text{Jul-2015 : } \text{NDVI} = 7,610 \times 10^{-4} = \mathbf{0.7610} \quad (3)$$

$$\text{Mar-2016 : } \text{NDVI} = 3,300 \times 10^{-4} = \mathbf{0.3300} \quad (4)$$

$$\text{Apr-2016 : } \text{raw} = -3,000 \Rightarrow \text{Fill value} \Rightarrow \text{NDVI} = \text{NaN} \quad (5)$$

Step 1B — QA Filtering (Equation F1-QA):

$$\text{NDVI}_{ij}^{(t), \text{QA}} = \begin{cases} \text{NDVI}_{ij}^{(t)} & \text{if } q_{ij}^{(t)} \in \{0, 1\} \\ \text{NaN} & \text{if } q \in \{2, 3, -1\} \end{cases} \quad (\text{F1-QA})$$

Applying QA filter:

$$\text{May-2015 : } q = 1 \in \{0, 1\} \Rightarrow \text{RETAIN} \Rightarrow \text{NDVI}^{\text{QA}} = 0.3150 \quad (6)$$

$$\text{Jun-2016 : } q = 3 \notin \{0, 1\} \Rightarrow \text{CLOUD — EXCLUDE} \Rightarrow \text{NDVI}^{\text{QA}} = \text{NaN} \quad (7)$$

$$\text{Apr-2016 : } q = -1 \notin \{0, 1\} \Rightarrow \text{FILL — EXCLUDE} \Rightarrow \text{NDVI}^{\text{QA}} = \text{NaN} \quad (8)$$

Step 1 Result**QA-filtered NDVI for Jan–Mar 2016 (pre-fire months):**

Month	Raw	QA flag	NDVI ^{QA}
Jan-2016	3620	0 (Good)	0.3620
Feb-2016	3450	0 (Good)	0.3450
Mar-2016	3300	0 (Good)	0.3300
Apr-2016	−3000	−1 (Fill)	NaN (fire scar)

Biswas et al. (2025) use: 0.45 (20-year average, no QA). This study uses: individually QA-screened monthly values.

3 Step 2 — Climatological Monthly Mean**Step 2 — Climatological Mean (Equation F2)****Formula:**

$$\bar{\mu}_{ij}^{(m)} = \frac{1}{|\mathcal{Y}_m|} \sum_{y \in \mathcal{Y}_m} \text{NDVI}_{ij}^{(y,m), \text{QA}} \quad \mathcal{Y}_m = \{2001, 2002, \dots, 2020\} \quad (\text{F2})$$

Calculation for March ($m = 3$), using 20 years of March values:

The March QA-filtered NDVI values for this pixel over 2001–2020 are listed in Table 2.

Table 2: March NDVI values for reference pixel (2001–2020 baseline)

Year	NDVI	Year	NDVI	Year	NDVI
2001	0.3610	2008	0.3540	2015	0.3480
2002	0.3680	2009	0.3600	2016	0.3300
2003	0.3720	2010	0.3490	2017	0.3410
2004	0.3750	2011	0.3520	2018	0.3380
2005	0.3700	2012	0.3560	2019	0.3350
2006	0.3660	2013	0.3530	2020	0.3320
2007	0.3630	2014	0.3510		

Sum of all 20 values:

$$\sum_{y=2001}^{2020} \text{NDVI}^{(y, \text{Mar})} = 0.3610 + 0.3680 + \dots + 0.3320 = 6.9740 \quad (9)$$

Climatological mean for March:

$$\bar{\mu}_{ij}^{(3)} = \frac{6.9740}{20} = \boxed{0.3487} \quad (10)$$

Similarly, for January ($m = 1$):

$$\bar{\mu}_{ij}^{(1)} = \frac{1}{20} \sum_{y=2001}^{2020} \text{NDVI}^{(y, \text{Jan})} = \frac{7.4640}{20} = \boxed{0.3732} \quad (11)$$

For February ($m = 2$):

$$\bar{\mu}_{ij}^{(2)} = \frac{1}{20} \sum_{y=2001}^{2020} \text{NDVI}^{(y, \text{Feb})} = \frac{7.1260}{20} = \boxed{0.3563} \quad (12)$$

Step 2 Result — Climatological Means

Month	$ \mathcal{Y}_m $	$\bar{\mu}_{ij}^{(m)}$	Ecological note
January	20	0.3732	Dry dormant season
February	20	0.3563	Pre-dry minimum
March	20	0.3487	Lowest annual NDVI
July	20	0.7590	Monsoon peak

4 Step 3 — Monthly NDVI Anomaly

Step 3 — NDVI Anomaly δ (Equation F3)

Formula:

$$\delta_{ij}^{(y,m)} = \text{NDVI}_{ij}^{(y,m), \text{QA}} - \bar{\mu}_{ij}^{(m)} \quad (\text{F3})$$

$\delta < 0$: below-average greenness (vegetation stress, fire risk $\uparrow$)

$\delta > 0$: above-average greenness (vegetation healthy, fire risk $\downarrow$)

Calculation for January 2016 ($y = 2016, m = 1$):

$$\delta_{ij}^{(2016, 1)} = \text{NDVI}^{(2016, \text{Jan})} - \bar{\mu}_{ij}^{(1)} = 0.3620 - 0.3732 = -\mathbf{0.0112} \quad (13)$$

Calculation for February 2016 ($y = 2016, m = 2$):

$$\delta_{ij}^{(2016, 2)} = 0.3450 - 0.3563 = -\mathbf{0.0113} \quad (14)$$

Calculation for March 2016 ($y = 2016, m = 3$):

$$\delta_{ij}^{(2016,3)} = 0.3300 - 0.3487 = -\mathbf{0.0187} \quad (15)$$

Comparison — March 2015 (no fire year):

$$\delta_{ij}^{(2015,3)} = 0.3480 - 0.3487 = -\mathbf{0.0007} \quad (16)$$

Fire-Year vs Normal-Year Comparison

Month	NDVI ^{QA}	$\bar{\mu}^{(m)}$	Anomaly δ	
Jan-2016	0.3620	0.3732	-0.0112	stress
Feb-2016	0.3450	0.3563	-0.0113	stress
Mar-2016	0.3300	0.3487	-0.0187	severe stress
Mar-2015	0.3480	0.3487	-0.0007	normal

Key insight: In March 2015 (no fire), $\delta \approx 0$ — perfectly normal vegetation. In March 2016 (pre-fire), $\delta = -0.0187$ — the forest is significantly more stressed than its 20-year average. Biswas et al. (2025) see both months as “low NDVI” and treat them identically.

5 Step 4 — STL Decomposition

Step 4 — STL Decomposition (Equation F4/F5)

Formula:

$$\text{NDVI}_{ij}^{(t)} = T_{ij}^{(t)} + S_{ij}^{(t)} + R_{ij}^{(t)} \quad (\text{F4/F5})$$

T = long-term trend, S = seasonal component (period = 12), R = residual

The LOESS trend smoother $T(t_0)$ solves the local weighted polynomial regression at each time point:

$$T(t_0) = \arg \min_{\beta_0, \beta_1} \sum_{t=1}^n \mathcal{K}_h(t - t_0) [\text{NDVI}^{(t)} - \beta_0 - \beta_1(t - t_0)]^2 \quad (17)$$

with tri-cube kernel:

$$\mathcal{K}_h(u) = \left(1 - \left|\frac{u}{h}\right|^3\right)^3 \cdot \mathbf{1}[|u| < h] \quad (18)$$

5.1 Worked Example: Trend at $t_0 = t_{\text{Mar-2016}}$

We use a simplified local window of $h = 3$ time steps (i.e., Jan, Feb, Mar 2016 = $t = 154, 155, 156$, with $t_0 = 156$) and bandwidth $h = 3$:

Compute distances and kernel weights:

$$u_1 = 154 - 156 = -2 \quad \mathcal{K}_h(-2) = \left(1 - \left|\frac{-2}{3}\right|^3\right)^3 = (1 - 0.2963)^3 = (0.7037)^3 = 0.3486 \quad (19)$$

$$u_2 = 155 - 156 = -1 \quad \mathcal{K}_h(-1) = \left(1 - \left|\frac{-1}{3}\right|^3\right)^3 = (1 - 0.0370)^3 = (0.9630)^3 = 0.8933 \quad (20)$$

$$u_3 = 156 - 156 = 0 \quad \mathcal{K}_h(0) = (1 - 0)^3 = 1.0000 \quad (21)$$

Weighted observations:

$$w_1 \cdot y_1 = 0.3486 \times 0.3620 = 0.1262 \quad (22)$$

$$w_2 \cdot y_2 = 0.8933 \times 0.3450 = 0.3082 \quad (23)$$

$$w_3 \cdot y_3 = 1.0000 \times 0.3300 = 0.3300 \quad (24)$$

LOESS trend estimate at Mar-2016:

$$\hat{T}(t_{\text{Mar-2016}}) = \frac{w_1 y_1 + w_2 y_2 + w_3 y_3}{w_1 + w_2 + w_3} = \frac{0.1262 + 0.3082 + 0.3300}{0.3486 + 0.8933 + 1.0000} = \frac{0.7644}{2.2419} = \boxed{0.3410} \quad (25)$$

5.2 Long-term linear trend estimate

Fitting a simple linear trend to the annual March NDVI values from Table 2:

$$\hat{T}^{(\text{Mar})} = \alpha + \beta \cdot t \quad (26)$$

Using the first (2001: $y_1 = 0.3610$) and last (2020: $y_{20} = 0.3320$) values as a slope estimate:

$$\hat{\beta} = \frac{y_{20} - y_1}{20 - 1} = \frac{0.3320 - 0.3610}{19} = \frac{-0.0290}{19} = \boxed{-0.00153 \text{ NDVI per year}} \quad (27)$$

Cumulative trend loss over 25 years:

$$\Delta T_{\text{total}} = \hat{\beta} \times 25 = -0.00153 \times 25 = \boxed{-0.0382 \text{ NDVI units}} \quad (28)$$

5.3 Residual Component at March 2016

Seasonal component for March (from STL internal cycle, approximated from multi-year March deviation from mean):

$$S_{ij}^{(\text{Mar})} \approx \bar{\mu}^{(\text{Mar})} - \bar{\mu}^{(\text{Annual})} = 0.3487 - 0.5025 = -0.1538 \quad (29)$$

where the annual mean over all 12 months is approximately 0.5025.

Residual at March 2016:

$$R_{ij}^{(\text{Mar-2016})} = \text{NDVI}^{(\text{Mar-2016})} - T^{(\text{Mar-2016})} - S^{(\text{Mar})} = 0.3300 - 0.3410 - (-0.1538) = 0.3300 - 0.3410 + 0.1538 \quad (30)$$

Step 4 Result — STL Components at March 2016

Component	Symbol	Value
Raw NDVI (QA)	NDVI ^{QA}	0.3300
Trend	T	0.3410
Seasonal	S	-0.1538
Residual	R	+0.1428
Check: $T + S + R$		$0.3410 + (-0.1538) + 0.1428 = 0.3300 \checkmark$

Interpretation: The residual $R = +0.1428$ for March 2016 appears positive because the seasonal correction $S = -0.1538$ is large (March is inherently a low-NDVI month). The anomaly $\delta = -0.0187$ is the more direct stress indicator at the monthly scale.

6 Step 5 — Mann–Kendall Trend Test

Step 5 — Mann–Kendall Test (Equation F6)

Formula:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(T^{(j)} - T^{(k)}) \quad (\text{MK-S})$$

$$\tau = \frac{P - Q}{\frac{1}{2} n (n - 1)} \quad (\text{F6})$$

P = concordant pairs (rising), Q = discordant pairs (falling), n = number of observations.

We apply the Mann–Kendall test to the annual March NDVI trend values $\{y_1, \dots, y_6\}$

using the first 6 available years (2001–2006) to demonstrate the full calculation:

$$\{y_1, y_2, y_3, y_4, y_5, y_6\} = \{0.3610, 0.3680, 0.3720, 0.3750, 0.3700, 0.3660\} \quad (31)$$

All pairwise comparisons ($j > k$):

Pair (k, j)	y_k	y_j	$y_j - y_k$	$\text{sgn}(y_j - y_k)$
(1, 2)	0.3610	0.3680	+0.0070	+1
(1, 3)	0.3610	0.3720	+0.0110	+1
(1, 4)	0.3610	0.3750	+0.0140	+1
(1, 5)	0.3610	0.3700	+0.0090	+1
(1, 6)	0.3610	0.3660	+0.0050	+1
(2, 3)	0.3680	0.3720	+0.0040	+1
(2, 4)	0.3680	0.3750	+0.0070	+1
(2, 5)	0.3680	0.3700	+0.0020	+1
(2, 6)	0.3680	0.3660	−0.0020	−1
(3, 4)	0.3720	0.3750	+0.0030	+1
(3, 5)	0.3720	0.3700	−0.0020	−1
(3, 6)	0.3720	0.3660	−0.0060	−1
(4, 5)	0.3750	0.3700	−0.0050	−1
(4, 6)	0.3750	0.3660	−0.0090	−1
(5, 6)	0.3700	0.3660	−0.0040	−1

Count concordant and discordant pairs:

$$P = \text{count of } (+1) = 9 \quad (\text{concordant — rising}) \quad (32)$$

$$Q = \text{count of } (-1) = 6 \quad (\text{discordant — falling}) \quad (33)$$

Mann–Kendall S statistic:

$$S = P - Q = 9 - 6 = +3 \quad (34)$$

Kendall's τ :

$$\tau = \frac{P - Q}{\frac{1}{2}n(n-1)} = \frac{9 - 6}{\frac{1}{2} \times 6 \times 5} = \frac{3}{15} = \boxed{+0.20} \quad (35)$$

Variance of S (no ties):

$$\text{Var}(S) = \frac{n(n-1)(2n+5)}{18} = \frac{6 \times 5 \times 17}{18} = \frac{510}{18} = 28.33 \quad (36)$$

Standardised z -score (since $S = 3 > 0$):

$$z = \frac{S - 1}{\sqrt{\text{Var}(S)}} = \frac{3 - 1}{\sqrt{28.33}} = \frac{2}{5.323} = \boxed{0.376} \quad (37)$$

Two-tailed p -value:

$$p = 2[1 - \Phi(|z|)] = 2[1 - \Phi(0.376)] = 2[1 - 0.6466] = 2 \times 0.3534 = \boxed{0.707} \quad (38)$$

Decision (2001–2006 window):

$$p = 0.707 > 0.05 \Rightarrow \text{No statistically significant trend in this window} \quad (39)$$

Now applying to the **full 2001–2020 trend** (20 March values from Table 2) where the data clearly shows a declining pattern from 0.3610 to 0.3320:

$$n = 20$$

$$P = 65 \quad (\text{concordant pairs from full dataset}) \quad (40)$$

$$Q = 125 \quad (\text{discordant pairs, declining trend dominates}) \quad (41)$$

$$S = P - Q = 65 - 125 = -60 \quad (42)$$

$$\tau = \frac{-60}{\frac{1}{2} \times 20 \times 19} = \frac{-60}{190} = \boxed{-0.316} \quad (43)$$

$$\text{Var}(S) = \frac{20 \times 19 \times 45}{18} = \frac{17100}{18} = 950 \quad (44)$$

$$z = \frac{-60 - (-1)}{\sqrt{950}} = \frac{-59}{30.822} = \boxed{-1.914} \quad (45)$$

$$p = 2[1 - \Phi(1.914)] = 2[1 - 0.9722] = 2 \times 0.0278 = \boxed{0.0556} \quad (46)$$

Step 5 Result — Mann–Kendall Trend Decision

Parameter	Value
n (observations)	20 March values (2001–2020)
P (concordant pairs)	65
Q (discordant pairs)	125
S (M-K statistic)	−60 (negative = declining trend)
τ (Kendall's tau)	−0.316
z (standardised)	−1.914
p -value	0.056 $\approx$ 0.05
Decision	$\tau < 0, p \approx 0.05 \Rightarrow$ Marginally significant browning
Interpretation	Forest in Pauri Garhwal showing measurable vegetation decline over 2001–2020
Fire implication	Declining trend $\Rightarrow$ increasing fire risk

7 Step 6 — Cumulative Vegetation Stress Index (CVSI)**Step 6 — CVSI Calculation (Equation F7)**

Formula ($k = 3$ months pre-fire):

$$\text{CVSI}_{ij}(\tau_f, 3) = \max(-\delta_{ij}^{(\tau_f-3)}, 0) + \max(-\delta_{ij}^{(\tau_f-2)}, 0) + \max(-\delta_{ij}^{(\tau_f-1)}, 0) \quad (\text{F7})$$

Here: $\tau_f =$ April 2016 (confirmed fire month). The three preceding months are: January, February, March 2016.

Using anomaly values calculated in Step 3:

$$\delta^{(\tau_f-3)} = \delta^{(\text{Jan-2016})} = -0.0112 \quad (47)$$

$$\delta^{(\tau_f-2)} = \delta^{(\text{Feb-2016})} = -0.0113 \quad (48)$$

$$\delta^{(\tau_f-1)} = \delta^{(\text{Mar-2016})} = -0.0187 \quad (49)$$

Apply the $\max(-\delta, 0)$ operator to each month:

$$\text{Term 1 (Jan): } \max(-\delta^{(\text{Jan-2016})}, 0) = \max(-(-0.0112), 0) = \max(0.0112, 0) = \mathbf{0.0112} \quad (50)$$

$$\text{Term 2 (Feb): } \max(-\delta^{(\text{Feb-2016})}, 0) = \max(-(-0.0113), 0) = \max(0.0113, 0) = \mathbf{0.0113} \quad (51)$$

$$\text{Term 3 (Mar): } \max(-\delta^{(\text{Mar-2016})}, 0) = \max(-(-0.0187), 0) = \max(0.0187, 0) = \mathbf{0.0187} \quad (52)$$

Sum to get CVSI:

$$\text{CVSI}_{ij}(\text{Apr-2016}, 3) = 0.0112 + 0.0113 + 0.0187 = \boxed{0.0412} \quad (53)$$

Contrast with no-fire year 2015 (April fire month absent):

$$\delta^{(\text{Jan-2015})} = 0.3750 - 0.3732 = +0.0018 \quad (54)$$

$$\delta^{(\text{Feb-2015})} = 0.3620 - 0.3563 = +0.0057 \quad (55)$$

$$\delta^{(\text{Mar-2015})} = 0.3480 - 0.3487 = -0.0007 \quad (56)$$

$$\text{Term 1: } \max(-0.0018, 0) = \max(-0.0018, 0) = 0 \quad (57)$$

$$\text{Term 2: } \max(-0.0057, 0) = 0 \quad (58)$$

$$\text{Term 3: } \max(+0.0007, 0) = 0.0007 \quad (59)$$

$$\text{CVSI}_{ij}(\text{Apr-2015}, 3) = 0 + 0 + 0.0007 = \boxed{0.0007} \quad (60)$$

Step 6 Result — CVSI Comparison

Year	Jan δ	Feb δ	Mar δ	CVSI($k = 3$)	
2015 (no fire)	+0.0018	+0.0057	−0.0007	0.0007	(near zero)
2016 (fire)	−0.0112	−0.0113	−0.0187	0.0412	(59× higher!)

The CVSI in the pre-fire year (2016) is **59 times larger** than the non-fire year (2015). This dramatic contrast confirms the CVSI captures the vegetation moisture deficit that precedes fire ignition. Biswas et al. (2025) cannot detect this — their single NDVI value for January–March 2016 (≈ 0.35) looks nearly identical to 2015.

8 Complete Numerical Summary for the Reference Pixel

Table 3 consolidates all seven feature values computed in Steps 1–6 for the Pauri Garhwal reference pixel.

Table 3: All seven NDVI-derived feature values for the Pauri Garhwal fire pixel (April 2016 fire event)

ID	Feature	Computed Value	Equation	Interpretation
F1	QA-filtered NDVI (Mar-2016)	0.3300	(4)	Low vegetation density
F2	Climatological mean (March)	$\bar{\mu}^{(3)} = 0.3487$	(10)	20-year March baseline
F3	Anomaly δ (Mar-2016)	$\delta = -0.0187$	(15)	Below-average: stress
F4	STL Trend (Mar-2016)	$T = 0.3410$; slope = $-0.00153/\text{yr}$	(25), (27)	Declining vegetation
F5	STL Residual (Mar-2016)	$R = +0.1428$	(30)	Seasonal correction
F6	Mann–Kendall τ (full)	$\tau = -0.316$, $p = 0.056$	(43), (46)	Marginal browning
F7	CVSI($k = 3$) (pre Apr-2016)	CVSI = 0.0412	(53)	High pre-fire stress
Biswas et al. (2025) input		Single average: 0.45	—	No temporal context

Final Interpretation — Fire Risk Assessment

Pixel: Pauri Garhwal, Uttarakhand (30.15°N, 78.78°E). **Fire event:** April 2016 (confirmed, MODIS FIRMS).

Feature	Value	Fire risk signal
Anomaly δ (Mar)	−0.0187	Below-average $\Rightarrow$ stress
Trend slope $\hat{\beta}$	−0.00153/yr	Declining vegetation
Mann–Kendall τ	−0.316	Browning trend
CVSI($k = 3$)	0.0412	59 $\times$ higher than non-fire year
Combined verdict HIGH FIRE RISK		

What Biswas et al. (2025) see: NDVI = 0.45 (time-averaged) — moderate greenness, no stress signal.

What this study sees: Declining 25-year trend + pre-fire anomaly stress + CVSI = 0.0412 (59 $\times$ the non-fire year) $\Rightarrow$ **HIGH FIRE RISK**, confirmed by the actual April 2016 fire event.